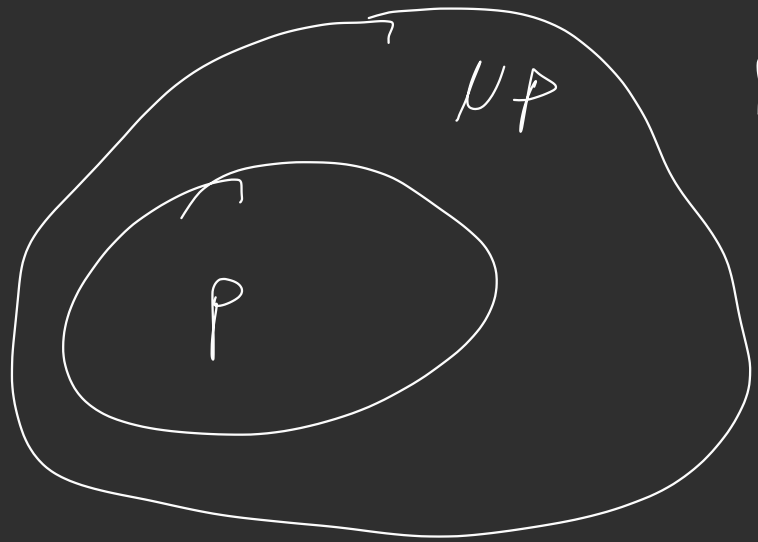

$P \subseteq NP$

$P \stackrel{?}{=} NP$

NP:

TSP, knapsack, Sudoku, crypto, SAT

P: easy to solve

poly. runtime on a DTM

NP: easy to check

poly. runtime on an NTM

SAT $\in$ NP:

logical syntax		boolean semantics
$\vee$	$\rightsquigarrow$	max
$\wedge$	$\rightsquigarrow$	min.
$\neg p$	$\rightsquigarrow$	$1-p$

<u>Log. form.:</u>

$\varphi = q \vee (\neg r \wedge \neg q) \in SAT$ ✓

<u>Q</u>: assignment?

$\mathcal{J}: Var = \{q, r\} \to \{0,1\}$ making φ true,

always true: $p \vee \neg p$

always false: $\underbrace{p \wedge \neg p}_{\notin SAT}$

pick $\mathcal{J}$ with $\mathcal{J}(q)=1$, $\mathcal{J}(r)=1$

$\rightsquigarrow \varphi^{\mathcal{J}} = \max(1, \min(0,0)) = \max(1,0) = 1$

Resolution:

input: CNF formula

can always transform, uses corresp.:

$PL_n \longleftrightarrow B_n$

$\varphi(p_1,\dots,p_n) \longmapsto \left(f_\varphi : \{0,1\}^n \to \{0,1\},\ \vec{b} \mapsto \varphi[b_1,\dots,b_n] \right)$

Ex.: $(p \vee \neg q) \wedge (r \vee q)$

$\underbrace{\neg p}\ \wedge\ \underbrace{(r \vee q)}\ \wedge\ \underbrace{(r \vee \neg s)}$

clauses

$\Updownarrow$

$\{\{\neg p\}, \{r, q\}, \{r, \neg s\}\}$

clause set represent.

Resolution: search for resolvent

$C_1 = \{p, \neg q, r\}$ $\qquad$ $C_2 = \{p, q, s, t\}$

resolvent $\{p, r, s, t\} = (C_1 \setminus \{\neg q\}) \cup (C_2 \setminus \{q\})$

only merge one var at a time

$\therefore$ Form. is not satisfiable iff we get empty clause $\square = \{\ \}$.